Definitions

1 Machine Learning

1.1 Hypothesis Space

The set of all functions a model can represent, denoted by $\mathcal{H}$.

$$\mathcal{H} = \{h : \mathcal{X} \to \mathcal{Y}\}$$

Here, h is a hypothesis mapping inputs $\mathcal{X}$ to outputs $\mathcal{Y}$. The actual hypothesis space contains only the functions allowed by the model.

1.2 Inductive Bias

The assumptions that guide a learning algorithm toward certain hypotheses, allowing it to generalize beyond its training data.

Restriction bias. Limits the hypothesis space $\mathcal{H}$ to a smaller set of possible functions.

$$\mathcal{H}_{\text{restricted}} \subseteq \mathcal{H}$$

A smaller hypothesis space can require less data to generalize, provided it contains a sufficiently good hypothesis.

Preference bias. Favors certain hypotheses over others without necessarily excluding them from $\mathcal{H}$.

For example, weight decay encourages solutions with smaller weights.